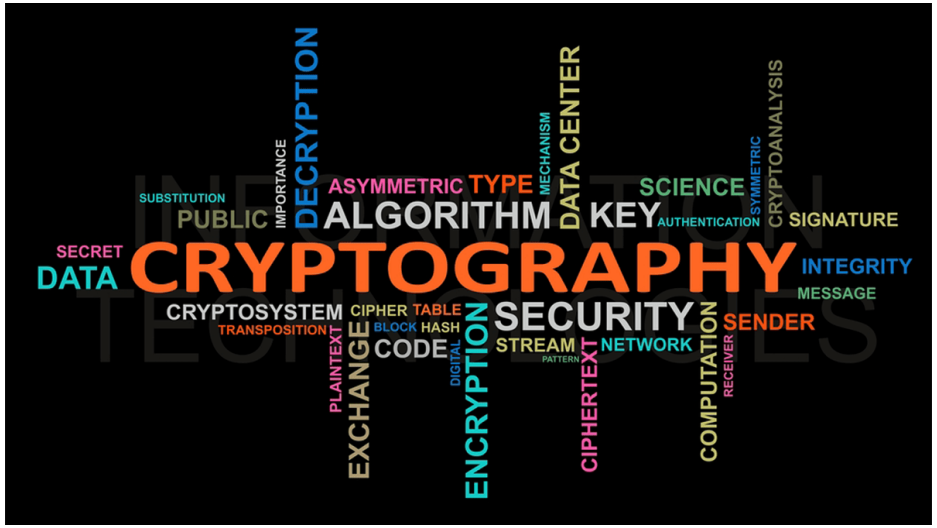


Cryptanalysis of PQC

FYP 1st Review

Group No. 22

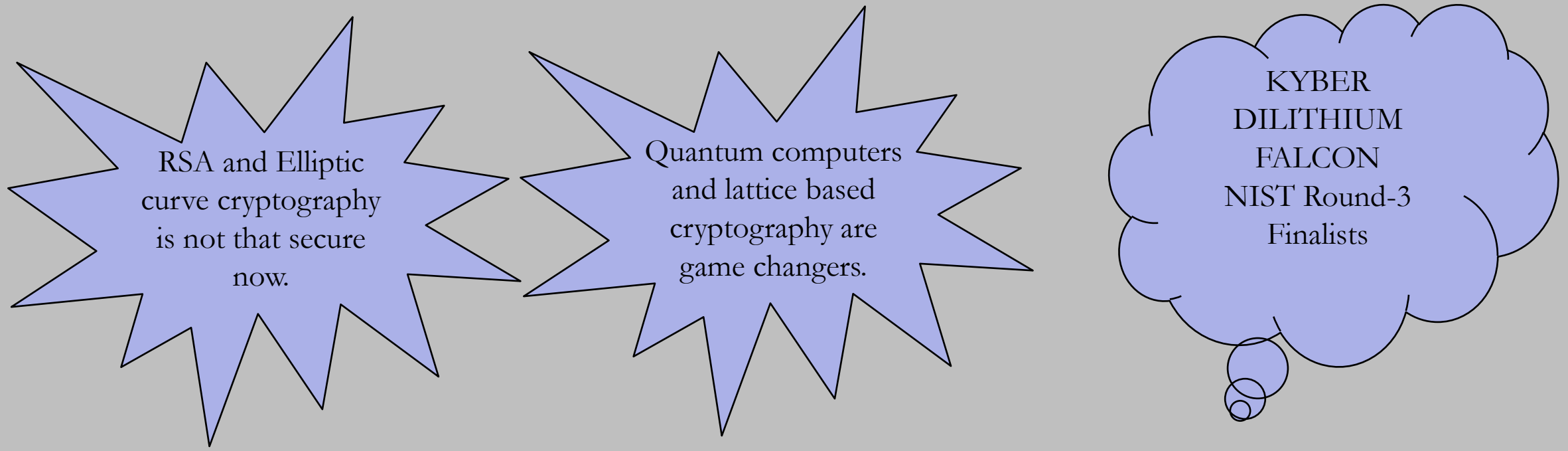
Guide: Kunwar Singh

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Problem Statement and Introduction

The rise of quantum computers poses a threat to traditional cryptographic schemes, making the search for quantum-resistant algorithms crucial. NIST has selected finalists for post-quantum cryptographic algorithms, but their security needs to be rigorously assessed. This project aims to cryptanalyze the NIST Round 3 finalists, identify potential vulnerabilities. This will inform the development of robust and secure post-quantum cryptographic systems for future use.

Existing Solutions

LWE — Learning with errors

- It's a lattice based asymmetric cryptosystem,
- Breaking it is as hard as solving lattice-hard problems.

Ring LWE — Modified LWE

- Asymmetric cryptosystem using LWE but over a monic polynomial ring (x^n+1).

KYBER and family

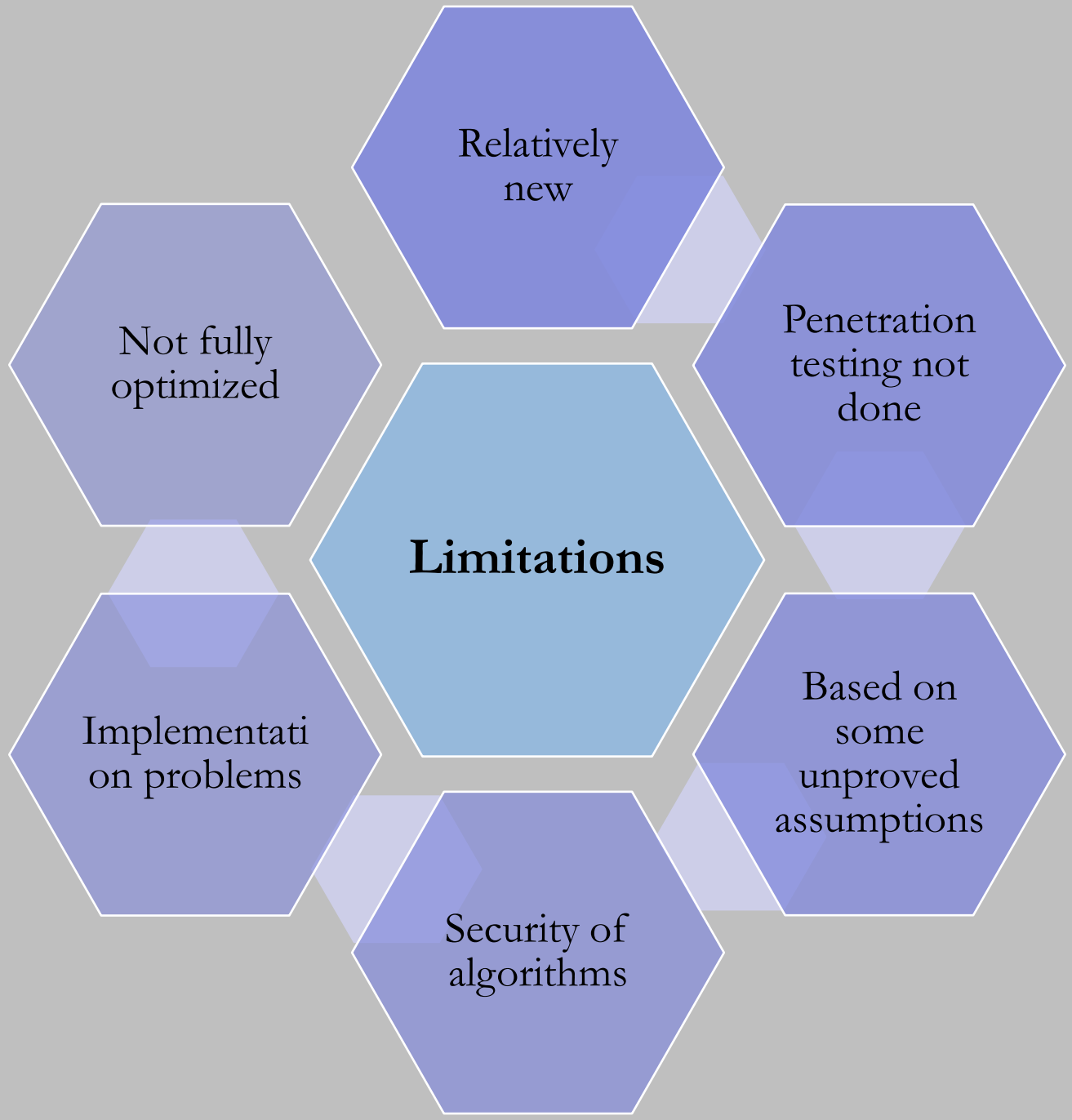
Kyber is an IND-CCA2-secure key encapsulation mechanism (KEM)

- Security is based on the hardness of solving the learning-with-errors (LWE) problem.
- Kyber is one of the finalists in the NIST post-quantum cryptography project.
- Specifically, Kyber-512 aims at security roughly equivalent to AES-128, Kyber-768 aims at security roughly equivalent to AES-192, and Kyber-1024 aims at security roughly equivalent to AES-256.

Other analysis

Cryptanalysis by other cryptanalysts

- Some of researchers and crypto engineer's have some specific implementation papers of algorithms.
- Some analysis on underlying mathematical problems.


Relatively new
Penetration testing not done
Based on some unproved assumptions
Security of algorithms
Implementation problems
Not fully optimized
Limitations

After NIST released it's 3 finalist of PQC algorithms, the discussion and heat around post quantum cryptography is increased. Something new always feels attractive so here we are.

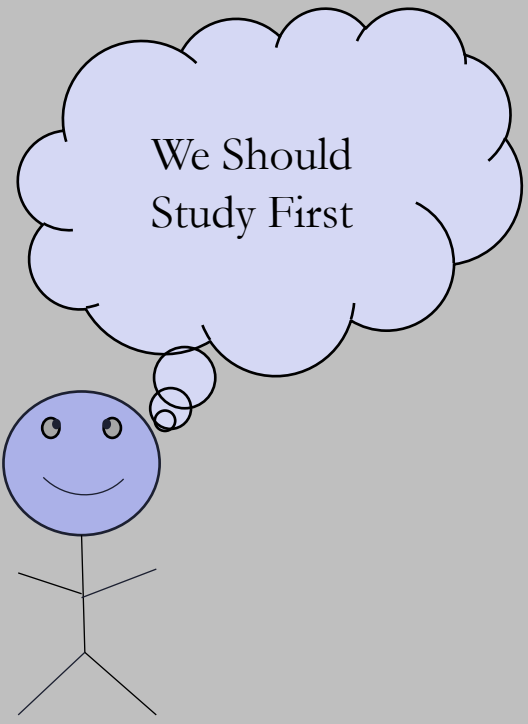
+

- Heavy mathematics.
- Contributing to a bigger cause.
- Working towards one of the future goal.
- CTF's are part of life so working towards problems in it is always fun.

=

Motivation

As a cryptanalyst, we need to understand the mathematical problems that underlie post-quantum cryptography, such as lattice-based, code-based, and multivariate-based cryptographic systems.



Lattices

LWE

BKZ

LLL

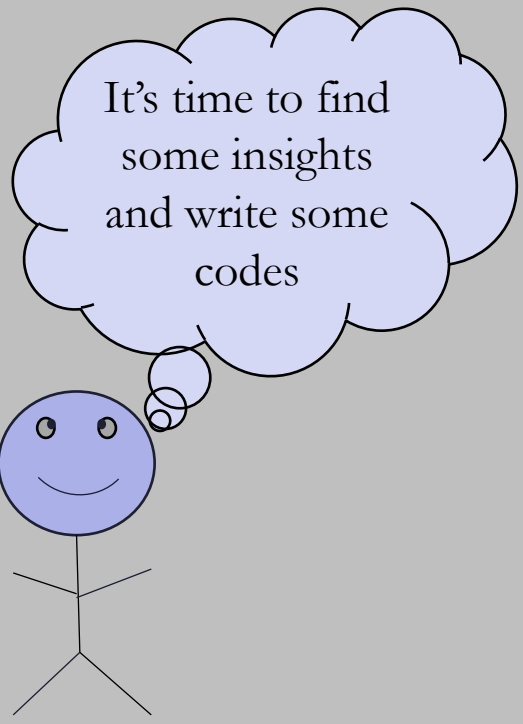
Shor's Algorithm

Grover's Algorithm

Differential cryptanalysis

Methodology

Analyse the security assumptions and investigate about hardness of a mathematical underlying problem, Identify potential vulnerabilities, checking weakness in key generation process and encryption/decryption.



- Effectiveness of a PQC algorithm.
  Test PQC algorithms on various crypto systems.
- Scope of optimizations in PQC algorithms.
  Detailed analysis on security, hardness, special cases, resources.
- Work towards proving or disproving of assumptions.
- Practical use cases.

Methodology

Post-quantum cryptography is a relatively new area of research, and there are not many established attack strategies for these systems. Try to develop new attack strategies that are specifically tailored to the post-quantum cryptographic system.



- This field is very new and have scope of lot of things, as an end result we are expecting that we will be able to find and prove some crucial points.
- We will be able to contribute to NIST's research and it will play a crucial role in penetration testing.
- Work towards publication of work and contribution in various fields.

Methodology

Time Schedule

Week-1
Study about LWE

Week-2
Algorithms of NIST

Week-3
Research on hardness of mathematical problems

Week-4
Implement algorithms

Week-5
Test algorithms on various cryptosystems

Week-6
Check scope of optimization and vulnerabilities

Week-7
Organize all codes, and research work

Week-8
Work on thesis and demo

Week-9
Work towards publication

Tools and Techniques













References

Post-quantum Cryptography: An Introduction (Shweta Agrawal) Link

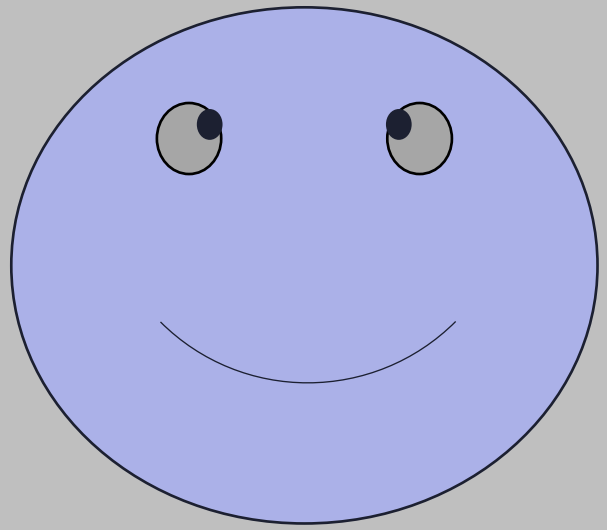
Lattice-based Cryptography (Daniele Micciancio and Oded Regev)Link

CRYSTALS – Kyber: a CCA-secure module-lattice-based KEM Link

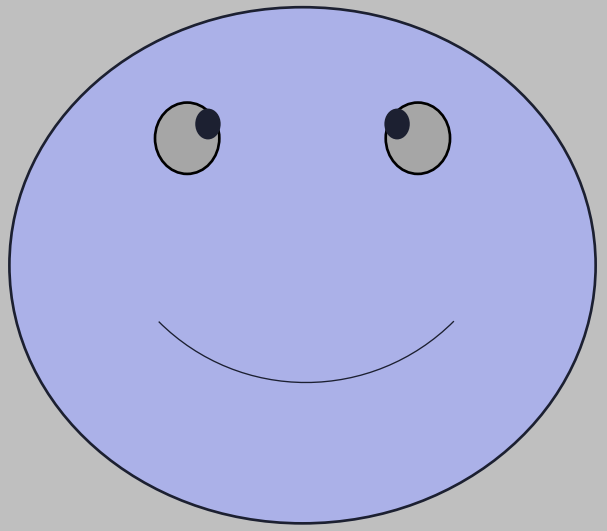
CRYSTALS-Kyber Algorithm Specifications And Supporting Documentation (version 3.01)Link

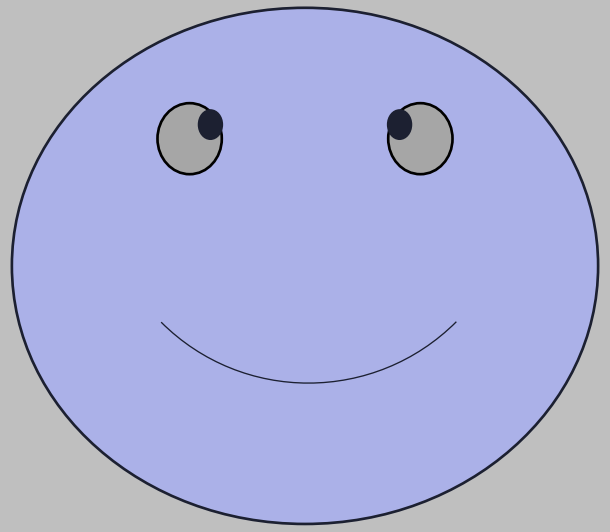
MATZOV. (2022). Report on the Security of LWE: Improved Dual Lattice Attack. Zenodo. Link

Mariano, Artur & Laarhoven, Thijs & Correia, Fabio & Rodrigues, Manuel & Falcao, Gabriel. (2017). A Practical View of the State-of-the-Art of Lattice-Based Cryptanalysis. IEEE Access. PP. 1-1. 10.1109/ACCESS.2017.2748179.



Thank You

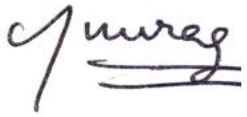




We promise that we will work hard



Dr. Kunwar Singh







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